

Membership Inference Attacks on Masked Diffusion Language Models

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Model: Qwen3-0.6B-diffusion-mdlm-v0.1 Benchmark: MIMIR (6 domains, 12,000 samples)

How Masked Diffusion LLMs Work: The Forward Process

At noise level t , every token is independently replaced by [M] with probability $(1 - \bar{\alpha}_t)$.

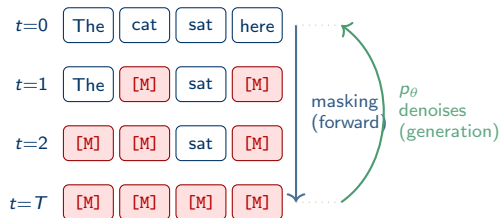
The marginal is:

$$q(\mathbf{x}_t \mid \mathbf{x}_0) = \bar{\alpha}_t \delta(\mathbf{x}_t = \mathbf{x}_0) + (1 - \bar{\alpha}_t) \delta(\mathbf{x}_t = [\text{M}])$$

where $\bar{\alpha}_t = \prod_{i=1}^t \alpha_i \downarrow 0$ as t grows.

Once a token becomes [M] it stays [M] — this is an absorbing Markov chain, not Gaussian noise.

Unlike autoregressive LLMs, there is no per-token log-probability. The model sees a partially masked sentence and predicts the hidden tokens.



How Masked Diffusion LLMs Work: Training and Inference

Training objective

A transformer θ learns to reconstruct masked tokens from their context. The objective is the Evidence Lower Bound (ELBO):

$$\mathcal{L}(\theta) = \mathbb{E}_{t, \mathbf{x}_t \sim q(\cdot | \mathbf{x}_0)} [\log p_{\theta}(\mathbf{x}_0 | \mathbf{x}_t)]$$

This is a variational lower bound on the data log-likelihood.

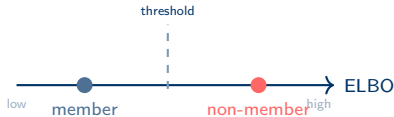
Fine-tuning on private data $\mathcal{D}_{\text{train}}$ continues this exact same ELBO objective on the new corpus.

Inference: start from a fully masked sequence $[[M]]^n$, then iteratively unmask tokens by sampling from p_{θ} until the sequence is fully recovered.

ELBO gap after fine-tuning

Member texts have **lower** ELBO than non-members after fine-tuning. The model reconstructs them more accurately because it has seen them during training.

This ELBO gap is the core membership signal we exploit.



The Privacy Problem

Membership Inference Attack (MIA)

Setup: given white-box access to a fine-tuned MDLM and a text sample x , determine whether x was in the fine-tuning data.

Prior work: SAMA (Chen et al., 2026) introduced the first MIA designed for discrete diffusion models. It uses subset-aggregated loss differences and requires only grey-box access (output loss, no internals).

Our approach

We extract a **46-dimensional feature vector** from the model's denoising trajectory and train a binary classifier.

Features from 11 signal groups measured at 4 noise levels ($\alpha \in \{5\%, 20\%, 35\%, 50\%\}$):

Signal group	What it captures
ELBO trajectory	loss at each α
Entropy traj.	prediction certainty
Mask consistency	multi-mask agreement
Hidden-state norms	layer activations
Attention patterns	cross-layer dynamics
Cross-model cosine	fine-tuned vs base

Experiment 1: Proof of Concept

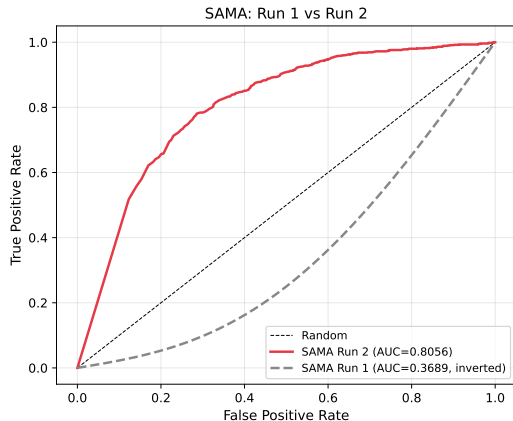
Data: MIMIR GitHub — 300 members, 300 non-members.

Setup: fine-tune Qwen3-0.6B-MDLM for 15 epochs to induce strong memorisation (loss: 3.86 $\rightarrow$ 1.59 nats). Extract the full **112-dimensional** feature vector using all 11 signal groups at $T = 10$ timesteps plus the cross-model cosine similarity.

SAMA result: AUC = 0.369 — below random chance. The attack is designed for moderate memorisation; heavy overfitting inverts its signal entirely. This revealed a design mismatch, not just an implementation bug.

Classifier result: XGBoost on all 112 features achieves AUC = 0.989, TPR @ 1% FPR = 88.9%.

White-box trajectory features are highly effective even in this small-sample setting where SAMA completely fails



SAMA AUC across Experiments 1 and 2, showing the design-mismatch issue and its resolution after adjusting the fine-tuning regime.

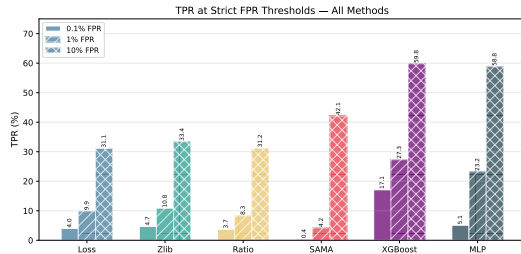
Experiment 2: Feature Refinement

Same setting: GitHub data, same model.

We reduced overfitting by using 5 epochs instead of 15, bringing SAMA back into its intended operating regime. We then refined the feature vector down to **46 dimensions** ($T = 4$ timesteps instead of 10) while retaining all 11 signal groups:

- ELBO values at $T = 4$ noise levels
- Prediction entropy at each noise level
- Mask consistency across 8 random masking patterns
- Hidden-state norms and cross-layer cosine similarity
- Attention entropy, cross-layer attention (AttenMIA-inspired)
- Cross-model cosine (fine-tuned vs base model)

The ELBO trajectory alone (4 numbers) outperforms SAMA on most metrics — pointing to it as the



TPR across FPR thresholds on GitHub: our classifier separates cleanly from SAMA and the black-box baselines.

Experiment 3: Full Evaluation

Scale-up: 6 MIMIR domains, 12,000 samples.

Domain	Text type
ArXiv	scientific papers
GitHub	source code
HackerNews	news and discussion
PubMed Central	biomedical abstracts
Pile-CC	CommonCrawl web text
Wikipedia	encyclopedic text

Per domain: fine-tune Qwen3-0.6B-MDLM on 1,000 members (5 epochs, $\text{lr} = 10^{-4}$), evaluate on 1,000 members and 1,000 non-members using the 46-feature pipeline from Experiment 2.

Attack classes compared:

Black-box baselines

Loss, Zlib normalisation, Ratio (fine-tuned vs base model loss). Access: output loss only.

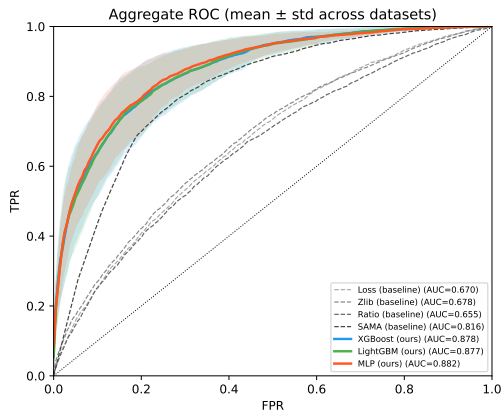
Grey-box: SAMA (corrected)

Subset-aggregated masked-loss difference.

White-box: our classifier

XGBoost and MLP trained on 46 trajectory features. 5-fold cross-validation with 95% CI on AUC.

Results: AUC-ROC Across All Domains



Mean ROC across 6 domains. Shaded band is ± 1 std.

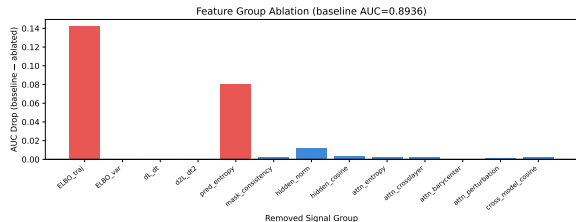
Domain	XGBoost	SAMA	Gap
Pile-CC	0.930	0.885	+0.045
HackerNews	0.928	0.833	+0.095
Wikipedia	0.892	0.823	+0.069
ArXiv	0.877	0.783	+0.094
PubMed	0.842	0.777	+0.065
GitHub	0.801	0.795	+0.006

TPR at 1% FPR:

- XGBoost: **15–39%** across domains
- SAMA: 3–10%
- **4–8 $\times$ improvement** at this threshold

GitHub is hardest: source code has high structural regularity across members and non-members, making the membership signal noisier despite having the largest ELBO gap of any domain.

Ablation: What Drives Detection?



Leave-one-group-out AUC drop, averaged across 6 domains. Larger drop = group matters more.

Leave one feature group out, retrain, measure AUC drop.

Feature group	Mean AUC drop
ELBO trajectory	+0.142
Predictive entropy	+0.081
Hidden-state norms	+0.012
Mask consistency	+0.002
Attention features	≈ 0

- ELBO trajectory dominates — removing it alone drops AUC by 0.142
- Prediction entropy adds a secondary but real signal (+0.081)
- Attention features are redundant once ELBO and entropy are present
- Just 8 features (ELBO + entropy, 4 each) capture most of the discriminative power

Why is low-noise ELBO so informative? The next

Privacy Theory: Wei et al. (2023)

Wei et al. derive per-instance differential privacy (pDP) guarantees for discrete denoising diffusion models.

Their transition: uniform noise over vocabulary of size K :

$$Q_t = \alpha_t I + (1 - \alpha_t) \cdot \frac{\mathbf{1}\mathbf{1}^\top}{K}$$

Any token can become any other token with equal probability.

Per-instance DP (pDP): measures privacy leakage specifically for one record v^* given a training set of size $s + 1$. The question: how much does the model's output shift when v^* is included vs excluded?

MDLM transition (absorbing)

MDLMs use:

$$Q_t = \alpha_t I + (1 - \alpha_t) \mathbf{e}_{[M]} \mathbf{1}^\top$$

A token can only become $[M]$, not any arbitrary vocabulary token. Once masked, it stays masked — this is the absorbing state.

The Wei et al. framework is the right tool to reason about MDLM privacy, but their bound was derived for the uniform case and does not transfer directly to the absorbing transition.

Privacy Theory: The Bound

Theorem 1 (Wei et al., simplified):

$$\delta(v^*, \varepsilon) \lesssim \underbrace{\bar{R}_t^{n-1}}_{\text{noise factor}} \cdot \underbrace{\frac{m}{s\varepsilon}}_{\text{data factor}}$$

where $\bar{R}_t = \bar{\alpha}_t$ is the cumulative masking level, n is sequence length, m is generation length, and s is the number of other training records.

What the noise factor means:

- At **high noise** ($t \rightarrow T$): $\bar{\alpha}_t \rightarrow 0$, so $\bar{R}_t^{n-1} \rightarrow 0$. Leakage is near zero — masking hides the data.
- At **low noise** ($t \rightarrow 0$): $\bar{\alpha}_t \rightarrow 1$, so $\bar{R}_t^{n-1} \rightarrow 1$. Leakage reaches $O(m/s\varepsilon)$ — the model is close to the training data and membership leaks.

Why this connects to the ablation:

Our ELBO features are measured at low masking ratios ($\alpha \in \{5\%, 20\%, 35\%, 50\%\}$). At these levels $\bar{\alpha}_t$ is close to 1, so the noise factor $\bar{R}_t^{n-1}$ is near its maximum.

The theorem tells us this is exactly the regime of highest privacy leakage. Our features probe the model precisely where membership information is most exposed — which is why ELBO at low noise is the dominant signal.

The gap

Wei et al. assume uniform transitions. MDLMs use absorbing transitions. The bound does not yet apply directly.

Experiment 4 — LLaDA-8B

Apply the same pipeline to LLaDA-8B (8B parameters, $13\times$ larger). Does greater capacity change the attack surface? Do larger models memorise more precisely, or generalise enough to hide individual records?

Bound for Absorbing Transitions

Wei et al. use uniform noise:

$$Q_t = \alpha_t I + (1 - \alpha_t) \frac{\mathbf{1}\mathbf{1}^\top}{K}$$

MDLMs use absorbing noise:

$$Q_t = \alpha_t I + (1 - \alpha_t) \mathbf{e}_{[M]} \mathbf{1}^\top$$

Goal: derive the analogue of Theorem 1 for the absorbing case. Does concentrating noise on a single state give stronger or weaker privacy than spreading it uniformly?

Summary

- White-box features substantially outperform SAMA and all black-box baselines across all 6 domains.
- The ELBO trajectory (4 numbers) drives most of the classifier's performance. Attention features add nothing.
- MDLMs are more privacy-vulnerable than prior work suggested.
- Wei et al. (2023) explain why: low-noise ELBO is exactly the regime of maximum privacy leakage.

Thank you.

References I

- ① Y. Chen, K. Zhang, Y. Du, E. Stoppa, C. Fleming, A. Kundu, B. Ribeiro, and N. Li. **Membership Inference Attacks Against Fine-tuned Diffusion Language Models.** *ICLR 2026*. arXiv:2601.20125.
- ② R. Wei, E. Kreacic, H. Wang, H. Yin, E. Chien, V. K. Potluru, and P. Li. **On the Inherent Privacy Properties of Discrete Denoising Diffusion Models.** *arXiv:2310.15524*, 2023.
- ③ P. Zaree, M. A. Al Mamun, Y. Dong, I. Alouani, and N. Abu-Ghazaleh. **AttenMIA: LLM Membership Inference Attack through Attention Signals.** *arXiv:2601.18110*, 2026.
- ④ D. Duan, A. Suri, N. Miresghallah, T. Le, W. Shi, Y. Zhu, T. Shi, M. Ryabinin, J. Min, and Y. Tsvetkov. **Do Membership Inference Attacks Work on Large Language Models?** *arXiv:2402.07841*, 2024. (MIMIR benchmark)